Seorhin Choi Curriculum Vitae

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EDUCATION

M.S. in Physics — Expected 2027
Ewha Womans University, Seoul
GPA 4.15/4.3 (official)
3.96/4.0 (U.S. scale)

B.S. in Physics — 2025
B.A. in Philosophy (double majors)
Ewha Womans University, Seoul
GPA 3.76/4.3, 95% (official)
3.70/4.0 (U.S. scale)

High School Diploma — 2020
Chungnam Samsung Academy, Asan

RESEARCH EXPERIENCE

Institute for Basic Science – Center for Quantum Nanoscience (QNS) — 2025 – present

M.S. Researcher

Main project: *Machine-learning-assisted multiplet calculations for lanthanides on MgO*

- Performed multiplet calculations for Er atoms on MgO surfaces by introducing ligand-field model with partially occupied 5d configurations
- Optimized multiplet parameters using Bayesian fitting to ESR-STM and XAS/XMCD measurements
- Reproduced experimentally observed g-tensor anisotropy of Er adatoms on MgO
- Extended the modeling approach to Ho, Tm, and Dy systems

Selected Experimental Collaborations

- Contributed to experiments on valence-electron magnetism in lanthanide atoms on superconductors
  (at SOLEIL Synchrotron, France)
- Performing XAS/XMCD data analysis on Gd, Bi_2Pd, and Td on $NbSe_2$
- Participated in experiments on element-specific spin polarization in intercalated magnetic transition metal dichalcogenides
  (at SOLEIL Synchrotron, France)

Institute for Basic Science – Center for Quantum Nanoscience (QNS) 2024

Undergraduate Research Intern

- Assisted in investigations of YPc_2 molecular spin qubits on Cu(111) surfaces
- Learned surface preparation techniques including sputtering, annealing, and molecular deposition
- Participated in low-temperature UHV-STM/STS measurements and experimental data acquisition

Trapped Ion Quantum Computing Laboratory, Ewha W. Univ. 2023

Undergraduate Research Intern

- Developed RF pulse-controlled AOM and PID-based laser control systems for optical experiments

TEACHING EXPERIENCE

Teaching Assistant – General Physics Laboratory 2025 – present
Department of Physics, Ewha Womans University

- Guided undergraduate students in experimental setup, measurements, and data analysis
- Developed grading rubrics and evaluated laboratory reports

Private Tutor (Physics & Mathematics) 2020 – present

- <u>Mentored 30+ students</u> in high-school mathematics and physics
- Taught <u>linear algebra</u> based on *Friedberg, Insel & Spence*

Mathematics Instructor 2022
Lim math, Cheonan

- Taught middle school mathematics in small-group classroom settings

HONORS AND SCHOLARSHIPS

Graduated Magna Cum Laude, Ewha Womans University 2025

Outstanding Ewha Scientist Scholarship, Ewha Womans University 2025 - 2026

PUBLCATIONS

Yaowu Liu*, Dasom Choi, Stefano Reale, Jeongmin Oh, **Seorhin Choi**, Fang Lei, We-hyo Seo, Andreas J. Heinrich, Soo-hyon Phark, and Fabio Donati, “All-electrical Coherent Control of a Single 4f Electron Spin”, manuscript in preparation

CONFERENCE PRESENTATIONS

"Identifying the quantum level structure of Er atoms on MgO using machine learning-assisted multiplet calculations", KPS Spring Meeting, Daejeon, South Korea, April 2026 (Oral Presentation)

"Identifying the quantum level structure of Er atoms on MgO using machine learning-assisted multiplet calculations", International Conference on Advanced Electromaterials (ICAE 2025), Jeju, South Korea, November 2025 (Poster session)

Accepted presentation "From States to Paths: Anyon Braiding and the Limits of State-Based Analysis". Korean Society for Philosophy of Science Annual Meeting (Upcoming), July 2026

TECHNICAL SKILLS

Programming & Data Analysis:
MATLAB, IGOR Pro, Bayesian optimization

Computational Physics:
Multiplet calculations (Quanty)

Experimental Techniques:
XAS/XMCD data analysis, synchrotron spectroscopy experiments

Outreach & Science Communication

Conceived and developed *STM Monopoly*, an educational outreach game introducing scanning tunneling microscopy; featured at a Harvard – MIT open house event and presented at the 2025 Korean Society for Science Education meeting

Initiated game-based physics outreach activities within the center, contributing to the development of the *Science Arcade* outreach program

Designed and developed *Schrodinger's Cat Party,* an educational game introducing concepts in quantum mechanics

Session Moderator, Women in Science, IBS conference on STM'25

LANGUAGES

Korean: Native
English: Advanced

REFERENCES

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